Exact Wave-Function Solution of Completely Free Rectangular Plates: Four-Corner Kirchhoff Jump and Mode Screening

Garrek McCune[a,*], Daniel Stutts[a]
[a] *Mechanical and Aerospace Engineering,*
Missouri University of Science and Technology, Rolla, MO 65409, USA
[*] Corresponding author. E-mail: ghmkfh@umsystem.edu

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Abstract

The Seok–Tiersten–Scarton wave-function method for rectangular plates is extended from clamped-free to completely free (FFFF). Closed-form wave functions satisfy the plate equation exactly; natural frequencies are zeros of a finite edge-determinant. For flexure the Kirchhoff twisting-moment jump at a free–free corner is not optional: a naive four-corner sum vanishes by a parity identity, and the physical jump is the checkerboard $(T-\nu)(-2)(\mathrm{TT}-\mathrm{TB}-\mathrm{WT}+\mathrm{WB})$. A frozen persistence screen (Screen B) separates physical modes from a small family of known artifacts. Independent half-model SHELL281 calculations at five aspect ratios $\ell/b = 1.0, 1.5, 2.0, 2.5, 3.0$, mesh-converged to $< 0.02\%$, confirm the screened candidates, typically inside 1%. Two low symmetric modes appear only on the persist basis (0.00% miss); one square-plate target ($\Lambda_{\mathrm{FE}} = 1.982$) is a named unmatched curiosity. Fifteen of Leissa's F–F–F–F Ritz entries agree with the matched roots within 1.10%. Identity-weighted MAC against the same U_Z eigenvectors confirms 36 of the 56 published flexural rows (every first symmetric mode MAC ≥ 0.959); Screen B leaks stay below the bar except where a leak frequency coincides with a physical mode. In-plane FFFF assembly has no Kirchhoff jump. Every non-rigid half-model PLANE183 target in $\bar{\Omega} \in [0.02, 2.50]$ is matched (92 unique frequencies, 16 of them recovered at an enlarged basis in a window centred on the finite-element value). Bardell, Langley and Dunsdon's first six F–F–F–F frequencies at $a/b = 1, 2$ agree to at most 1.69%, and nineteen of Gorman's superposition eigenvalues to at most 1.72% (independent finite elements versus those two sources at most 0.019% and 0.11%). No eigenvector cross-check is presented for the in-plane family. Default clamped-free assembly is unchanged.

Keywords: rectangular plate; free vibration; exact solution; free boundary; Kirchhoff corner; spurious modes; modal assurance.

1 Introduction

Seok, Tiersten and Scarton [3, 4] gave exact-wave-function solutions for cantilevered rectangular plates, out-of-plane (Part 1) and in-plane (Part 2). The method is the rectangular counterpart of their annular-sector papers [1, 2]: analytic wave functions in one direction, a variational statement on the remaining pair of edges, natural frequencies as zeros of a small characteristic determinant. Those papers stop at clamped-free. Completely free plates are the harder extension, because every edge is natural and Kirchhoff theory requires a twisting-moment jump at each free–free corner.

A companion paper [10] treats the annular-sector FFFF problem, including the spurious-zero discrimination instruments that two weakly enforced circumferential edges demand. The rectangular problem is not a transplant of that work. The axial solution is closed-form rather than Frobenius; the contour that produces the corner jump is a rectangle with four free corners rather than a sector with two; and the artifact mechanism that must be screened is a mixture of basis-limited dips and a few ℓ/b-independent leak families, not the annular residual-screen taxonomy.

This paper (i) re-derives the four-corner flexural jump from Part 1 Eq. (16) on a closed contour; (ii) shows why a same-sign four-term sum is identically zero and why the surviving combination is a checkerboard; (iii) defines a persistence screen that does not retune a threshold to chase FE; (iv) reports FFFF flexural tables at five aspect ratios against independent half-model finite elements, against Leissa's F–F–F–F Ritz values, and against the same FE eigenvectors via MAC; (v) extends the in-plane assembler to four free edges (no corner jump) and compares with FE at the same five aspect ratios and with Bardell, Langley and Dunsdon [5] at $a/b = 1, 2$ and Gorman [7] at $a/b = 1, 2$.

The rectangular cantilever tables of Parts 1 and 2 are already reproduced by the same package (default clamped-free assembly); that path is not modified here.

2 Related work

Flexural (out-of-plane) vibration of rectangular plates under every combination of clamped, simply supported, and free edges was tabulated comprehensively by Leissa using the Rayleigh–Ritz method [6]; the completely free (FFFF) entries in that table remain the standard reference point for this configuration, independent of and roughly contemporaneous with the Seok–Tiersten–Scarton cantilever line [3, 4] extended here. For in-plane (extensional) motion, Bardell, Langley and Dunsdon's hierarchical finite-element treatment already supplies one independent free-free comparison at $a/b = 1, 2$ [5], checked directly in §5.3; Gorman's superposition-method solution, built from beam eigenfunctions rather than a polynomial or finite-element basis, gives a second, independently derived in-plane free-free frequency set [7]. Both lines of Ritz/superposition work converge to the exact frequencies as their basis is enlarged, but neither is itself exact: each satisfies the governing equation only in an energy-averaged sense over an assumed displacement field.

The method used here differs from all three. Extending Seok, Tiersten and Scarton's cantilever wave-function solution [3, 4] to the completely free case, §3–§5 build closed-form solutions of the plate equation directly and locate natural frequencies as zeros of a finite characteristic determinant, with the free–free flexural case additionally requiring the four-corner Kirchhoff jump derived in §3.2. Convergence here is a property of the truncated wave-function series entering that determinant, not of an assumed global displacement field, so the finite-element checks of §4 and §5.2 and the Bardell comparison of §5.3 test the present method against independently-derived reference values rather than one approximate method against another of the same kind. Direct numerical comparisons against Leissa's F–F–F–F table and Gorman's in-plane superposition values are in §4.5 and §5.4.

3 Flexural formulation

3.1 Geometry and determinant

A rectangular plate of half-length ℓ and half-breadth b, aspect ratio ℓ/b, thickness H, isotropic Kirchhoff flexure, Poisson ratio $\nu = 0.30$. The Seok/Tiersten/Scarton nondimensional frequency Λ is used throughout. Finite-element comparisons convert ANSYS [8] cyclic frequencies by the factor

$K = 24.6644\,\mathrm{Hz}$ per unit Λ, from the closed-form factor $K = (\pi H)/(8b^2)\sqrt{G/[6\rho(1-\nu)]}$ with $G = E/[2(1+\nu)]$ and the geometry and material of §4 ($b = 1\,\mathrm{m}$). Hertz are converted in Python with this K; the decks' own printed Λ_{FE} column is not used (the APDL square-root is missing at runtime).

Even (SYM) and odd (ANTI) classes are the half-model mirror conditions: SYM $\leftrightarrow$ $UY = ROTX = ROTZ = 0$ on the cut; ANTI $\leftrightarrow$ $UX = UZ = ROTY = 0$. Production bases are $n_{\mathrm{real}} = 6$, $n_{\mathrm{cpair}} = 0$ (SYM) and 5/1 (ANTI). The persist screen uses three conjugate pairs on both classes.

3.2 Four-corner jump

Part 1 Eq. (16)'s last integral is the twisting-moment contribution $\varphi = n_a t_{ab} s_b$ around a contour with outward normal, oriented counterclockwise. At a free–free corner the integrand jumps. The four corners of the rectangle are not equivalent under the local (n, s) frame: TT and WB are even under the global product of edge tangents; TB and WT pick up a Jacobian -1. The physical jump is therefore the checkerboard

$$\mathrm{corner}_{ff} = (T-\nu)(-2)\,(\mathrm{TT} - \mathrm{TB} - \mathrm{WT} + \mathrm{WB}). \tag{1}$$

A naive sum of four cantilever-style terms, $p_{1t}q_{0t} + p_{1b}q_{0b}$ and cyclic, vanishes identically because

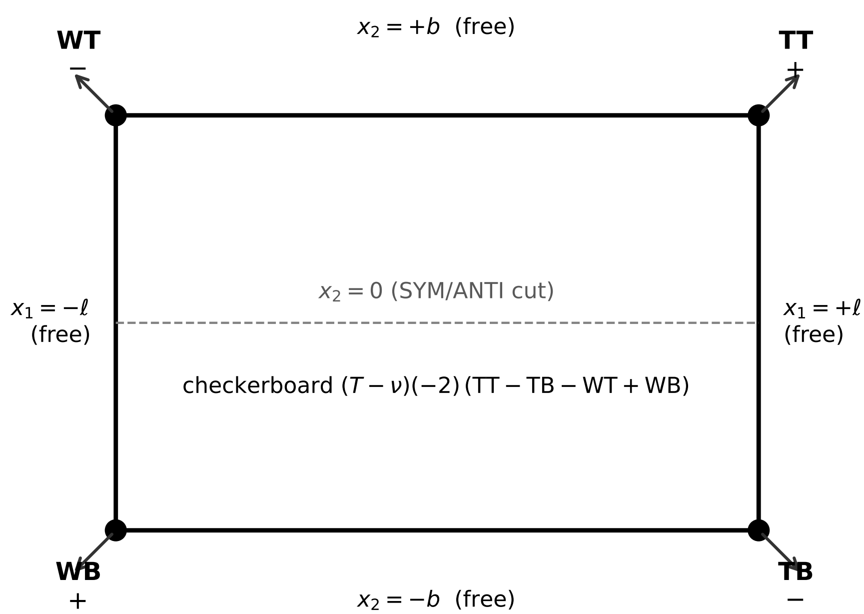

Figure 1: Completely free rectangle with outward normals at the four corners. TT and WB enter the jump with a plus, TB and WT with a minus: the checkerboard of Eq. (1). The dashed line is the half-model cut.

those two products sum to zero (parity identity). Under that identity the checkerboard is twice the three-corner single-point combination that had been used as an underived stand-in. The default cantilever assembly (three free edges, one clamped) is a different contour and is not altered.

The line-by-line contour trace, the four-corner table, the parity identity in both classes, and the algebraic check that the checkerboard is exactly twice the single-point candidate are Supplementary Material, §S.1.

3.3 Screen B

Discovery scans $\sigma_{\min}(\Lambda)$ on the production basis. Every listed dip with $\sigma < 0.3$ is re-scanned on the persist basis ($n_{\text{cpair}} = 3$, window ± 0.02, step 0.002). A dip **persists** if a persist-basis local minimum remains within 0.01 in Λ. That cut was frozen before the extra-ℓ/b geometries were run and is not retuned. The same rule, with $\bar{\Omega}$ in place of Λ and the in-plane persist basis of §5.2, is applied to the extensional candidates; what the rectangle has no analogue of is the annular companion's *residual* screen, not persistence itself.

Screen B kills a recurring antisymmetric pair near $\Lambda = 0.04$ and 0.46, and labels a symmetric leak at 0.41 that persists but does not match FE. It does not kill a deep extra near $\Lambda = 1.65$ (tiny σ, no FE partner) or a shallow symmetric extra near 1.92. Those are reported as false positives, not published modes.

Two genuine symmetric FE modes ($\ell/b = 2.0$, $\Lambda = 0.543$; $\ell/b = 3.0$, $\Lambda = 0.670$) never appear as production-basis dips. On the persist basis they sit on the FE values at 0.00% miss. Production discovery is therefore a floor, not a census: those two modes are recovered by the persist screen, not by silently enlarging the published detector.

A post-hoc sensitivity check replaces the fixed 0.002-step persist scan with a golden-section continuum minimisation at the same persist basis. Two Table 2 entries – $\ell/b = 1.5$ SYM $\Lambda^* = 5.380$ and $\ell/b = 2.5$ ANTI $\Lambda^* = 6.260$ – sit at $|\Delta\Lambda| = 0.0103$ and 0.0110 at that continuum optimum, outside the frozen 0.01 cut by 3–10%, though within it under the fixed-step scan that was actually run to discover them. The closest extensional case ($\ell/b = 3.0$ ANTI $\bar{\Omega}^* = 2.2400$) stays inside at $|\Delta\bar{\Omega}| = 0.0094$. Screen B's own margin is genuinely thin for those two flexural entries; independent FE agreement is not (0.03% and 0.65% miss respectively, Table 2). The cut is not retuned; the two entries are retained on the strength of that independent FE match, and neither meets the shape bar of §4.3.

4 Flexural finite-element comparison

Half-model SHELL281, $E = 210\,\text{GPa}$, $\nu = 0.30$, $\rho = 7800\,\text{kg}\,\text{m}^{-3}$, $H = 0.04\,\text{m}$, $b = 1\,\text{m}$. Mesh 1 uses element size $b/20$ along the half-breadth; mesh 2 uses $b/30$. Divisions along the length scale as NDIVX = round($2(\ell/b) \times 20$) and $\times 30$. Mesh 1 versus mesh 2 agrees to $< 0.02\%$ on the older decks and 0.001% on $\ell/b = 2.0$ and 3.0. Rigid-body (near-zero Hz) entries are discarded. Cross-parity matches are not counted. At $\ell/b = 1.0$ (square plate) some SYM and ANTI FE frequencies coincide; that degeneracy is physical.[1]

A refined candidate is a **MATCH** if the production-basis local minimum after a ± 0.04, step-0.002 polish lies within 3% of a same-parity FE Λ, and Screen B still passes. Double-dips that share one FE partner count once (closer kept). High-Λ discovery uses a 5% coarse cut, then drops machine zeros at 4.000 and 6.000 ($\sigma \sim 10^{-18}$ to 10^{-22}) and collapses doubles; the tables below keep the 3% subset unless noted.

4.1 Primary refined matches ($\Lambda \lesssim 2.4$)

Table 1 reports all 26 primary refined flexural matches with $\Lambda \lesssim 2.4$ across the five aspect ratios.

[1] Job-level detail: Supplementary Material, §S.4.

Table 1: Primary refined flexural matches ($\Lambda \lesssim 2.4$) against half-model SHELL281. Persist-basis-only rows are marked; they never appear as production-basis dips.

ℓ/b	class	Λ^*	Λ_{FE}	miss	note
1.0	ANTI	1.366	1.35288	0.97%	
1.5	SYM	0.964	0.96347	0.06%	1st SYM
1.5	SYM	2.254	2.24373	0.46%	
1.5	ANTI	0.906	0.89838	0.85%	1st ANTI
1.5	ANTI	2.124	2.07036	2.59%	weakest primary
2.0	SYM	0.5434	0.54338	0.00%	persist basis only
2.0	SYM	1.510	1.50735	0.18%	
2.0	SYM	2.234	2.22558	0.38%	
2.0	ANTI	0.674	0.66858	0.81%	1st ANTI
2.0	ANTI	1.482	1.47069	0.77%	
2.5	SYM	0.348	0.34766	0.10%	1st SYM
2.5	SYM	0.978	0.96559	1.29%	
2.5	SYM	1.888	1.88370	0.23%	
2.5	SYM	2.278	2.27059	0.33%	
2.5	ANTI	0.534	0.53126	0.52%	1st ANTI
2.5	ANTI	1.148	1.14003	0.70%	
2.5	ANTI	1.918	1.90392	0.74%	
3.0	SYM	0.242	0.24127	0.30%	1st SYM
3.0	SYM	0.6698	0.66984	0.00%	persist basis only
3.0	SYM	1.320	1.31785	0.16%	
3.0	SYM	2.162	2.15417	0.36%	
3.0	SYM	2.256	2.24841	0.34%	
3.0	ANTI	0.444	0.44044	0.81%	1st ANTI; 0.46-family coincidence
3.0	ANTI	0.936	0.93044	0.60%	
3.0	ANTI	1.528	1.51698	0.73%	
3.0	ANTI	2.258	2.24305	0.67%	

First-mode travel, quoting Table 1's Λ^*: SYM $0.964 \to 0.5434 \to 0.348 \to 0.242$ as ℓ/b goes $1.5 \to 2.0 \to 2.5 \to 3.0$ (the square plate's first SYM is the unmatched $\Lambda_{\text{FE}} = 1.982$ of §4.4); ANTI $1.366 \to 0.906 \to 0.674 \to 0.534 \to 0.444$ as ℓ/b goes $1.0 \to 1.5 \to 2.0 \to 2.5 \to 3.0$. At $\ell/b = 3.0$ the historical ANTI 0.46 leak sits on the real first ANTI; it is published as that mode in Table 1, not as evidence that the leak is physical at other aspect ratios.

4.2 Higher flexural matches

Table 2 extends the comparison past that cutoff with 30 further flexural matches at miss $\leq 3\%$, giving 56 published flexural rows in all.

Table 2: Selected higher flexural matches with miss $\leq 3\%$. ‡Persist-basis distance exceeds the frozen Screen B cut at the continuum optimum (§3.3); retained on FE agreement. Neither row meets the shape bar of §4.3.

ℓ/b	class	Λ^*	Λ_{FE}	miss
1.0	SYM	2.460	2.45439	0.23%
1.0	SYM	3.530	3.49250	1.07%
1.0	SYM	6.190	6.16026	0.48%
1.0	SYM	6.460	6.36553	1.48%
1.0	ANTI	3.520	3.49250	0.79%
1.5	SYM	5.380‡	5.38152	0.03%
1.5	ANTI	3.860	3.83354	0.69%
2.0	SYM	3.000	2.99812	0.06%
2.0	SYM	3.650	3.62227	0.77%
2.0	ANTI	2.580	2.55180	1.11%
2.0	ANTI	4.060	4.02613	0.84%
2.0	ANTI	6.200	6.17866	0.35%
2.5	SYM	3.190	3.19522	0.16%
2.5	SYM	4.180	4.13819	1.01%
2.5	SYM	5.410	5.34819	1.16%
2.5	ANTI	2.920	2.89206	0.97%
2.5	ANTI	4.180	4.14962	0.73%
2.5	ANTI	5.720	5.67275	0.83%
2.5	ANTI	6.260‡	6.21965	0.65%
2.5	ANTI	6.400	6.36131	0.61%
3.0	SYM	2.480	2.45935	0.84%
3.0	SYM	3.340	3.32528	0.44%
3.0	SYM	3.660	3.62467	0.97%
3.0	SYM	5.660	5.59746	1.12%
3.0	SYM	6.180	6.14744	0.53%
3.0	ANTI	3.160	3.14253	0.56%
3.0	ANTI	4.260	4.23529	0.58%
3.0	ANTI	5.560	5.51872	0.75%
3.0	ANTI	6.220	6.18191	0.62%
3.0	ANTI	6.380	6.34339	0.58%

4.3 Eigenvector cross-check (MAC)

Frequency agreement does not by itself identify a mode. For each published Λ^* the analytical w is reconstructed from the null vector of the free-free characteristic matrix and scored against the same-parity half-model SHELL281 U_Z eigenvector at the matching Λ_{FE}, using the identity-weighted MAC $|\langle u_Z, w\rangle|^2/(\|u_Z\|^2\|w\|^2)$ of Allemang [9], the same form used in the annular companion [10]. The bar 0.744 is that paper's production call (the midpoint of its $\nu = 0.30$ extensional calibration window), applied here unre-tuned as a conservative shape-confirmation threshold, not as a new calibration against these data. FE hertz are converted in Python with $K = 24.6644$; the decks' own printed Λ column is not used.

Of the 56 published flexural rows of Tables 1–2, 36 meet the bar: 25 of 29 symmetric and 11 of 27 antisymmetric. The other 20 are frequency matches only and no shape is claimed for them.

Isolated primary matches confirm: every first symmetric mode has MAC ≥ 0.959, and four of the five first antisymmetric modes lie at or above the bar (the $\ell/b = 2.5$ first ANTI saturates at 0.655 across a fine Λ grid, so the frequency match stands and the shape is reported as ambiguous). The bar's placement is not delicate: no row lies between 0.655, the highest MAC reached by an excluded row, and 0.751, the lowest confirmed one, so the confirmed set is the same for any threshold in that interval. Known Screen B leaks split: the SYM 0.41 family has MAC ≤ 0.05 against every same-parity FE mode; the ANTI 0.46 family stays below the 0.744 bar (MAC 0.19–0.43) except at $\ell/b = 3.0$, where it coincides with the real first ANTI and scores 1.000 as that mode. At $\ell/b = 2.5$ a 1% FE pair (3.16376 and 3.19522) is frequency-nearest-matched to $\Lambda^* = 3.190 \to 3.19522$; the production-basis shape attaches to 3.16376, and an $n_{\mathrm{cpair}} = 6$ reconstruction returns MAC $= 0.915$ against 3.19522, so the published frequency partner is the one the enlarged basis selects. Higher-Λ reconstructions can land on a nearby kernel or on the known machine-zero at $\Lambda = 4, 6$; those frequency matches are not withdrawn. The confirmed rows and the reconstruction basis used for each are Supplementary Material, §S.5.[2] Figure 2 is the first symmetric mode at $\ell/b = 1.5$: production-basis w against the matching SHELL281 U_Z, MAC $= 1.000$. A dps$=40$ persist-basis golden-section polish of the seven Table 1 ANTI rows that do not meet the bar converges to the published Λ^* (max travel 8×10^{-4}) and does not lift persist-basis MAC across 0.744, so coarseness of Λ^* is not what holds those rows below it, and the bar is not retuned.

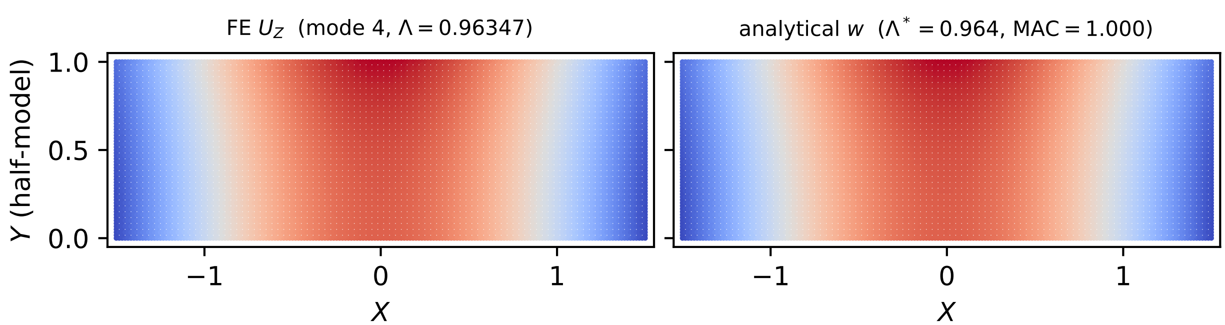

Figure 2: Half-model first symmetric flexural mode at $\ell/b = 1.5$. Left: SHELL281 U_Z (mode 4, $\Lambda_{\mathrm{FE}} = 0.96347$). Right: analytical w at $\Lambda^* = 0.964$, MAC $= 1.000$.

4.4 Open flexural items

- $\ell/b = 1.0$ SYM $\Lambda_{\mathrm{FE}} = 1.982$: unmatched, closed as a numerical curiosity. A persist-basis scan over $[1.73, 2.23]$ stays within one order of magnitude of its floor (ratio 3.12), with no dip near 1.982 and no tail toward the next SYM root at 2.454 (matched separately by $\Lambda = 2.460$). Not a missing mode; Screen B is not retuned.

[2] Job-level detail: Supplementary Material, §S.4.

- $\ell/b = 1.5$ ANTI $\Lambda^* = 2.124$ (Leissa $(3,2)$, miss 2.59%) is the weakest primary frequency match and does not MAC- confirm against FE#5 even at $n_{\mathrm{cpair}} = 6$. The frequency row is kept; the shape is not claimed.
- Several FE modes near 2.5–2.9 fail Screen B even when a production dip is within ∼1.5% (e.g. 2.0 SYM 2.622, 3.0 SYM 2.887, 1.5 SYM 2.588, 2.5 SYM 2.467). The screen is not retuned to collect them.
- Not published: SYM 0.41 leak; ANTI 0.04/0.46 except the $\ell/b = 3.0$ coincidence above; ANTI ∼1.65; shallow SYM ∼1.92; $\Lambda = 4, 6$ machine zeros.

4.5 Comparison with Leissa (1973)

Leissa's Rayleigh–Ritz survey of rectangular-plate vibration [6] tabulates F–F–F–F frequency parameters $\lambda = \omega a^2\sqrt{\rho h/D}$ (Table C15 of that reference) at five aspect ratios, three of which, $a/b = 1.0, 1.5, 2.5$, coincide with the flexural aspect ratios used here. Using $a = 2\ell$, $D = Eh^3/[12(1-\nu^2)]$ and the same $H = 0.04\,\mathrm{m}$, $E = 210\,\mathrm{GPa}$, $\rho = 7800\,\mathrm{kg\,m^{-3}}$ as §4, λ converts to Λ through $\Lambda = (\lambda/a^2)/(2\pi K\sqrt{\rho h/D})$, using the same $K = 24.6644$ Hz-per-unit-Λ constant as the finite-element comparison. The conversion is self-consistent to 0.05% on the one entry independently checkable against the published $\Lambda^* = 1.366$ ($\ell/b = 1.0$, ANTI): Leissa's fundamental $(2,2)$ mode, $\lambda = 13.489$, converts to $\Lambda = 1.3667$.

Leissa's own (m,n) half-wave counts fix the parity: odd n is y-symmetric, which is SYM here, and even n is y-antisymmetric, which is ANTI. Table 3 reports every Leissa entry in those three columns that converts within 2% of a *same-parity* Λ_{FE} already listed in Tables 1–2; pairing is by parity, not by nearest frequency.

Table 3: Flexural matches against Leissa (1973), Table C15, converted as above and paired by parity. The (m,n) column is Leissa's own half-wave count (odd n: y-symmetric = SYM here; even n: y-antisymmetric = ANTI). Miss is $|\Lambda^* - \Lambda_{\mathrm{Leissa}}|/\Lambda_{\mathrm{Leissa}}$, the same convention as Table 4.

ℓ/b	class	(m,n)	$\Lambda_{\mathrm{Leissa}}$	Λ^*	Λ_{FE}	miss
1.0	ANTI	$(2,2)$	1.3667	1.366	1.35288	0.05%
1.0	SYM	$(3,1)$	2.4755	2.460	2.45439	0.63%
1.0	ANTI	$(3,2)$	3.5487	3.520	3.49250	0.81%
1.0	SYM	$(2,3)$	3.5487	3.530	3.49250	0.53%
1.0	SYM	$(4,1)$	6.2339	6.190	6.16026	0.70%
1.5	ANTI	$(2,2)$	0.9064	0.906	0.89838	0.04%
1.5	SYM	$(3,1)$	0.9728	0.964	0.96347	0.91%
1.5	ANTI	$(3,2)$	2.1009	2.124	2.07036	1.10%
1.5	SYM	$(1,3)$	2.2648	2.254	2.24373	0.48%
2.5	SYM	$(3,1)$	0.3509	0.348	0.34766	0.82%
2.5	ANTI	$(2,2)$	0.5358	0.534	0.53126	0.33%
2.5	SYM	$(4,1)$	0.9749	0.978	0.96559	0.32%
2.5	ANTI	$(3,2)$	1.1589	1.148	1.14003	0.94%
2.5	SYM	$(5,1)$	1.9040	1.888	1.88370	0.84%
2.5	ANTI	$(4,2)$	1.9353	1.918	1.90392	0.89%

All fifteen conversions land within 1.10% of the matched root Λ^*, using no information from this paper's own solver beyond the conversion constant K; the independent finite-element values agree with the same fifteen entries to within 1.62%. The two eigenvalues Leissa lists at $\lambda = 35.024$ for the square plate reproduce the SYM/ANTI degeneracy this paper's own FE and solver both find there, and enter as the $(3,2)$ and $(2,3)$ rows.

Parity, rather than frequency, does real work at $\ell/b = 2.5$, where Leissa's $(5, 1)$ ($\lambda = 117.45$, y-symmetric) converts to $\Lambda = 1.9040$ and his $(4, 2)$ ($\lambda = 119.38$, y-antisymmetric) to 1.9353, while this paper's SYM and ANTI FE values sit at 1.88370 and 1.90392. Nearest-frequency matching would pair the y-symmetric $(5, 1)$ with the antisymmetric FE mode at 0.005% and leave $(4, 2)$ unaccounted for. Table 3 pairs each with its own parity instead: the agreement is 0.84% and 0.89% rather than 0.005%, and every row in the table is a same-parity comparison.

Three further Leissa entries in these three columns convert close to a finite-element mode that Tables 1–2 do not contain. Leissa's second $\ell/b = 1.0$ mode ($\lambda = 19.789$, whose quoted internal convergence check at 13.06% is the worst-behaved of that column) converts to $\Lambda = 2.0050$, 1.12% from the $\ell/b = 1.0$ SYM $\Lambda_{\mathrm{FE}} = 1.98249$ named in §4.4 as an unmatched curiosity: an independent Ritz solution also struggles with this mode, which weighs against it being an artifact of this paper's own screening. Leissa's $\ell/b = 1.5$ $\lambda = 58.201$ converts to $\Lambda = 2.6209$, 1.25% from the FE mode at 2.58809 named in §4.4 as failing Screen B despite a nearby production dip, so the literature value corroborates that a real mode sits there. His $\ell/b = 1.5$ $\lambda = 67.494$ converts to $\Lambda = 3.0394$, 1.17% from the FE mode at 3.00395, which likewise has no published match.

5 In-plane motion

5.1 No Kirchhoff corner jump

Seok, Tiersten and Scarton Part 2 Eq. (1) is Part 1 Eq. (16) with flexure set to zero: only $\int_{c_N} n_a t_{ab}^{(0)} \delta u_b \, ds$ and $\int_{c_C} u_b \delta(n_a t_{ab}^{(0)}) \, ds$ remain on the boundary. The twisting-moment exact differential that produces the out-of-plane corner jump is absent. After the long free edges are satisfied exactly, Eq. (36) is wall (clamped) plus tip (free traction) only. The in-plane assembler is that form ($\mathbf{K}$ = wall + tip); the class has no corner term. Completely free in-plane motion is the same two traction integrals on both short edges — still no jump. The free-free option is an edge-list swap, not a new formula. Default clamped-free assembly is bit-identical to the cantilever path.

5.2 Finite-element comparison

Half-model PLANE183, same geometry and material as §4, at $\ell/b = 1.0, 1.5, 2.0, 2.5, 3.0$. Mesh 1 versus mesh 2 agrees to 0.004%. Rigid-body (near-zero Hz) entries are discarded. The Seok/Tiersten/Scarton in-plane parameter is $\bar{\Omega} = \omega/\bar{\omega}$, $\bar{\omega} = (\pi/(2b))\sqrt{c_{66}/\rho}$, $c_{66} = E/[2(1 + \nu)]$. Finite-element comparisons convert ANSYS cyclic frequencies by $\bar{\Omega} = f/804.481\,\mathrm{Hz}$, independent of ℓ/b. SYM $\leftrightarrow$ $UY = 0$ on the cut; ANTI $\leftrightarrow$ $UX = 0$. Conversion constants and the full 92-row match list are Supplementary Material, §§S.2–S.3.

Production in-plane discovery uses the persist-sized basis ($n_{\mathrm{real}} = 5$, $n_{\mathrm{cpair}} = 3$) directly, not the cantilever default $n_{\mathrm{cpair}} = 0$: that default starves the FFFF spectrum (a band of real FE targets produced no dip at all until the basis was enlarged). A candidate is a **MATCH** if a same-basis local minimum after a ± 0.02, step-0.002 polish lies within 3% of a same-parity FE $\bar{\Omega}$. Sixteen FE targets that the $n_{\mathrm{cpair}} = 3$ scan left uncovered recover at $n_{\mathrm{cpair}} = 5$ in a ± 0.05 window centred on the FE value itself (15 at 0.00% miss, one at 0.14%). This is **not** Screen B and is not adopted as a portable residual screen. The annular companion's SUBDOM instrument [10] remains annular-IP-specific.

After collapsing double-dips onto the same FE partner, **every** one of the 92 unique non-rigid FE targets in $\bar{\Omega} \in [0.02, 2.50]$ ($5+4+8+6+10+7+13+10+16+13$ across the five aspect ratios and both parities) has a matched root; zero remain uncovered. Sixteen of the 92 are the enlarged-basis recoveries above, found in a window centred on the finite-element value rather than by blind discovery, and are tagged as such in the full list. First-mode travel of the matched $\bar{\Omega}^*$ is regular:

SYM 1.330 → 1.050 → 0.810 → 0.640 → 0.540 and ANTI 1.240 → 0.780 → 0.520 → 0.380 → 0.280 as ℓ/b goes 1.0 → 1.5 → 2.0 → 2.5 → 3.0.

No eigenvector cross-check is presented for the in-plane family. The resolved branch pool near several in-plane frequencies is thin — three to five candidates — and largely insensitive to the width of the branch search, so a reconstruction there would not carry the confirm/reject separation that makes the flexural comparison of §4.3 informative. The in-plane result in this paper is a frequency census against independent finite elements and against §5.3–§5.4; nothing is claimed about in-plane mode shapes.

5.3 Comparison with Bardell, Langley and Dunsdon (1996)

Bardell, Langley and Dunsdon [5] give the first six non-zero in-plane frequencies of isotropic F–F–F–F rectangles at $a/b = 1$ and 2, printed under Figure 1 mode plots rather than tabulated (their one table is a simply-supported convergence study). Their parameter is $\Omega_B = \omega a/C_L$ with $C_L^2 = E/[\rho(1-\nu^2)]$. The letter does not state ν numerically; their own Table 1, an exact S–S–S–S convergence study, gives the square-plate pair $(0,1) = (1,0) = 1.859$, which equals $\pi\sqrt{(1-\nu)/2} = 1.8586$ at $\nu = 0.30$, so the conversion uses that value.

The plates identify as $a = 2\ell$, so $a/b = \ell/b$ and

$$\Omega_B = \bar{\Omega} \cdot \pi \cdot (\ell/b) \cdot \sqrt{(1-\nu)/2}. \tag{2}$$

Figure 1's caption reads "(a) $a/b = 1$; (b) $a/b = 2$", but the (a) panel is drawn 2:1 with unique frequencies and the (b) panel is square with the repeated pair 2.472, 2.472. The body text assigns repeats to $a/b = 1$ and unique frequencies to $a/b = 2$. The mapping below follows the rendered proportions and that physics, not the caption letters (the annular companion [10] recorded the same defect; the source's own Table 1 does not have it). Supplementary Material §S.2 restates the mapping.

Table 4: Bardell, Langley and Dunsdon [5] F–F–F–F first six non-zero frequencies at $a/b = 1$ and 2, converted onto this paper's $\bar{\Omega}$ and paired with the matched root and independent FE. Miss is $|\bar{\Omega}^* - \Omega_B^{\text{conv}}|/\Omega_B^{\text{conv}}$. Aspect ratios are assigned by plate proportions (square + repeated $2.472 = a/b = 1$), not Figure 1's (a)/(b) caption letters.

ℓ/b	class	Bardell Ω_B	$\Omega_B \to \bar{\Omega}$	$\bar{\Omega}^*$	$\bar{\Omega}_{\text{FE}}$	miss
1.0	ANTI	2.321	1.24880	1.2400	1.24858	0.70%
1.0	SYM	2.472	1.33004	1.3300	1.32984	0.00%
1.0	ANTI	2.472	1.33004	1.3298	1.32984	0.02%
1.0	SYM	2.628	1.41397	1.4100	1.41422	0.28%
1.0	SYM	2.987	1.60713	1.6100	1.60733	0.18%
1.0	SYM	3.452	1.85732	1.8600	1.85745	0.14%
2.0	ANTI	1.954	0.52567	0.5200	0.52557	1.08%
2.0	SYM	2.961	0.79657	0.8100	0.79652	1.69%
2.0	ANTI	3.267	0.87889	0.8800	0.87891	0.13%
2.0	ANTI	4.726	1.27139	1.2715	1.27148	0.01%
2.0	ANTI	4.784	1.28700	1.2800	1.28703	0.54%
2.0	SYM	5.205	1.40025	1.4000	1.40011	0.02%

All twelve published F–F–F–F values are matched. Solver versus converted Bardell: max miss 1.69%, eight of twelve inside 0.3%. Independent FE versus converted Bardell: max miss 0.019%. The three largest solver misses sit on $n_{\text{cpair}} = 3$ roots that already passed the 3% FE cut on the refine grid (printed $\bar{\Omega}^*$ to 0.01); the two $n_{\text{cpair}} = 5$ mop-up roots that fall inside Bardell's first six sit at 0.01–0.02%. The repeated square-plate pair $2.472, 2.472$ is the SYM/ANTI degeneracy at $\bar{\Omega}_{\mathrm{FE}} = 1.32984$.

5.4 Comparison with Gorman (2004)

Gorman's superposition-method solution for in-plane vibration of the completely free rectangular plate [7] tabulates symmetric–symmetric, antisymmetric–antisymmetric, and mixed-parity mode families separately at the square plate and at one rectangular aspect ratio, $b/a = 0.5$, i.e. $a/b = 2$, coinciding with $\ell/b = 1.0$ and $\ell/b = 2.0$ here. Gorman's own tables already cross-check against Bardell's values directly (his "Ref. [1]"); comparing them against the present solver independently triangulates three separate methods.

One conversion subtlety was found and is recorded here for reproducibility. Gorman's tabulated λ^2 column is numerically *half* of the equivalent Bardell $\bar{\Omega}_B$ for the same mode (e.g. Gorman's $\lambda^2 = 1.160$ for the square plate's first symmetric–symmetric mode converts, after doubling, to $\Omega_B = 2.320$, matching Bardell's own published 2.321 for that mode to four figures) despite both papers quoting an apparently identical formula $\lambda^2 = \omega a\sqrt{\rho(1-\nu^2)/E}$: Gorman's quarter-plate schematic (his Fig. 2) evidently uses a as the *half*-length rather than the full edge length assumed by the nomenclature list. Doubling Gorman's tabulated values before applying Eq. (2) resolves the discrepancy to within FE-level agreement (below); using Gorman's values unconverted would have understated every frequency by very nearly a factor of two.

Table 5: In-plane matches against Gorman (2004), doubled and converted via Eq. (2). Miss is $|\bar{\Omega}^* - \bar{\Omega}_{\mathrm{Gorman}}|/\bar{\Omega}_{\mathrm{Gorman}}$, the same convention as Table 4; it is computed before $\bar{\Omega}_{\mathrm{Gorman}}$ is rounded to the four places shown. The class column is this paper's half-model parity, which is the opposite label to Gorman's quarter-plate family names (his symmetric–symmetric modes are ANTI here); pairing follows the finite-element parity, not his nomenclature.

ℓ/b	class	$\bar{\Omega}_{\mathrm{Gorman}}$	$\bar{\Omega}^*$	$\bar{\Omega}_{\mathrm{FE}}$	miss
1.0	ANTI	1.2483	1.2400	1.24858	0.66%
1.0	SYM/ANTI (degenerate)	1.3300	1.3300/1.3298	1.32984	0.02%
1.0	SYM	1.4140	1.4100	1.41422	0.28%
1.0	SYM	1.6077	1.6100	1.60733	0.14%
1.0	SYM	1.8573	1.8600	1.85745	0.14%
1.0	SYM/ANTI (degenerate)	2.0037	2.0000	2.00319	0.18%
1.0	ANTI	2.3168	2.3200	2.31523	0.14%
2.0	ANTI	0.5262	0.5200	0.52557	1.17%
2.0	SYM	0.7963	0.8100	0.79652	1.72%
2.0	ANTI	0.8792	0.8800	0.87891	0.10%
2.0	ANTI	1.2714	1.2715	1.27148	0.01%
2.0	ANTI	1.2870	1.2800	1.28703	0.54%
2.0	SYM	1.4011	1.4000	1.40011	0.08%
2.0	SYM	1.4145	1.4142	1.41422	0.02%
2.0	SYM	1.4446	1.4400	1.44333	0.32%
2.0	SYM	1.6539	1.6500	1.65355	0.24%
2.0	SYM	1.7745	1.7500	1.77390	1.38%
2.0	ANTI	1.8444	1.8400	1.84467	0.24%
2.0	ANTI	2.1382	2.1400	2.13652	0.09%

Nineteen of Gorman's twenty-four tabulated eigenvalues land inside the $\bar{\Omega} \in [0.02, 2.50]$ window covered by the full in-plane match list (Supplementary §S.3). All nineteen match a published root within 1.72%, and the independent finite-element values agree with the same nineteen to within 0.11%. The two largest solver misses, 1.72% and 1.38% at $\ell/b = 2.0$, are the same $n_{\mathrm{cpair}} = 3$ discovery roots printed to 0.01 that carry the largest Bardell misses; the FE column shows the underlying agreement is far tighter than the printed resolution of $\bar{\Omega}^*$. That the finite-element values reproduce Bardell to 0.019% and Gorman to 0.11% is consistent with the two being, by Gorman's own account, independent confirmations of each other. The remaining five Gorman eigenvalues, all at $\ell/b = 1.0$ (Gorman tabulates only four modes per family there, and the higher-parity ones run past $\bar{\Omega} = 2.5$), exceed the FE window's upper cutoff and are not compared.

6 Discussion

The four-corner jump is a derivation rather than a fit; Screen B is frozen; five aspect ratios of independent FE agree at the level the annular flexural tables claimed [10]. The remaining flexural issues are named misses and named artifacts.

The two papers split instruments as well as geometry. The annular companion screens a two-edge Galerkin projection by pointwise residuals (two-sided in flexure; one-sided in extension, closed by SUBDOM and by the MAC bar reused here). After the long edges of the rectangle are satisfied exactly, that residual taxonomy has nothing to act on: the remaining defect is a mixture of basis-limited dips and a few ℓ/b-independent leak families, which is why §3.3 uses persistence instead of a residual cut. What the rectangle has no analogue of is the residual screen itself, in either motion family: the persistence rule of §3.3 is applied to the flexural and extensional candidates alike, while SUBDOM stays annular-IP-specific. Completely free in-plane motion on the rectangle also has no Kirchhoff jump, and what its result rests on is the 92/92 frequency census together with Bardell and Gorman — a frequency claim, with no shape cross-check behind it.

What this paper does *not* claim: that production-basis flexural discovery is complete; that the in-plane matches are shape-confirmed; that the unre-tuned MAC bar is a new calibration on these data.

This paper is the rectangular companion of the annular manuscript, not an expansion of it. The annular contribution is the two-edge FFFF detector and its discrimination instruments on a sector. This paper's contribution is the four-free-edge rectangular jump, the in-plane edge-list extension that needs no jump, and the tables that follow from both.

7 Conclusions

The Seok–Tiersten–Scarton rectangular method extends to completely free flexure once the closed-contour Kirchhoff jump is assembled as a checkerboard rather than a vanishing four-term sum. A frozen complex-pair persistence screen, checked against half-model SHELL281 at $\ell/b = 1.0, 1.5, 2.0, 2.5, 3.0$, yields a set of even- and odd-parity frequencies that match FE typically inside 1%. Two symmetric modes require the persist basis to appear; one square-plate symmetric target is a named unmatched curiosity, not a missing mode. The same assembler extends to completely free in-plane motion with no corner term. Half-model PLANE183 at the five flexural aspect ratios matches every non-rigid target in $\bar{\Omega} \in [0.02, 2.50]$: 92 unique frequencies, sixteen of them recovered at an enlarged basis in a window centred on the finite-element value rather than by blind discovery.

Three independent literature sets bracket those tables. Bardell, Langley and Dunsdon's 1996 first six F–F–F–F frequencies at $a/b = 1, 2$, converted onto $\bar{\Omega}$, agree with the matched roots to at most 1.69%; fifteen entries of Leissa's F–F–F–F table agree with the flexural roots to at most 1.10%; nineteen of Gorman's superposition eigenvalues agree to at most 1.72%. Measured instead against the independent finite-element values, those same three sets agree to 0.019%, 1.62% and 0.11%.

Shape validation is flexural only. MAC against the half-model U_Z eigenvectors confirms 36 of the 56 published flexural rows, every first symmetric mode among them at MAC ≥ 0.959, and the known Screen B leaks stay below the bar except where a leak frequency coincides with a physical mode. No in-plane eigenvector comparison is presented. The elastic FFFF baseline on both the annular [10] and rectangular geometries is the intended foundation for piezoelectric constitutive coupling and for using the exact frequencies to identify those constants; that is later work, not a claim of this paper.

Data and code availability

The solver, finite-element decks, validated tables, and from-scratch regression suite are released under an MIT license as a public GitHub repository, `https://github.com/McCune1/plate_solver`. For the present paper that includes the free-free assembler, the Screen B implementation and every production probe behind Tables 1–5, the APDL decks and their frequency listings at all five aspect ratios, and regression gates for both free-free rectangular assemblers; the per-mode nodal eigenvector dumps behind §4.3 are roughly 88 MB and are regenerable from the shipped mode-extraction decks rather than tracked. The public tag `v1.0.0` matches the annular companion manuscript [10]; a manuscript-matching tag for the present rectangular paper will be added at submission. Tables here were captured under `SOLVER_VERSION 2026-07-10.s10`. Results under a different version should be checked against the clamped-free regression gates first. The four-corner derivation is Supplementary Material, §S.1; in-plane conversion constants and the 92-row match list are §§S.2–S.3; cluster job numbers are §S.4; the flexural MAC table is §S.5.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Garrek McCune: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Daniel Stutts:** Conceptualization, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used generative AI tools to assist with manuscript and code organization, drafting, editing, and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

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